

# SIMATS ENGINEERING

## ASSESSMENT TOOL 2

**COURSE CODE: MLA03**

**COURSE NAME: Reinforcement learning**

---

### COURSE OUTCOME COVERED

CO4: Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization. (BL3, BL4, BL5)

**Assessment Tool Weightage: 40%**

---

**Dr . DUMPALA PRASANTH**

**QUESTIONS: 25**

**Total Marks: 50**

### Annexure A

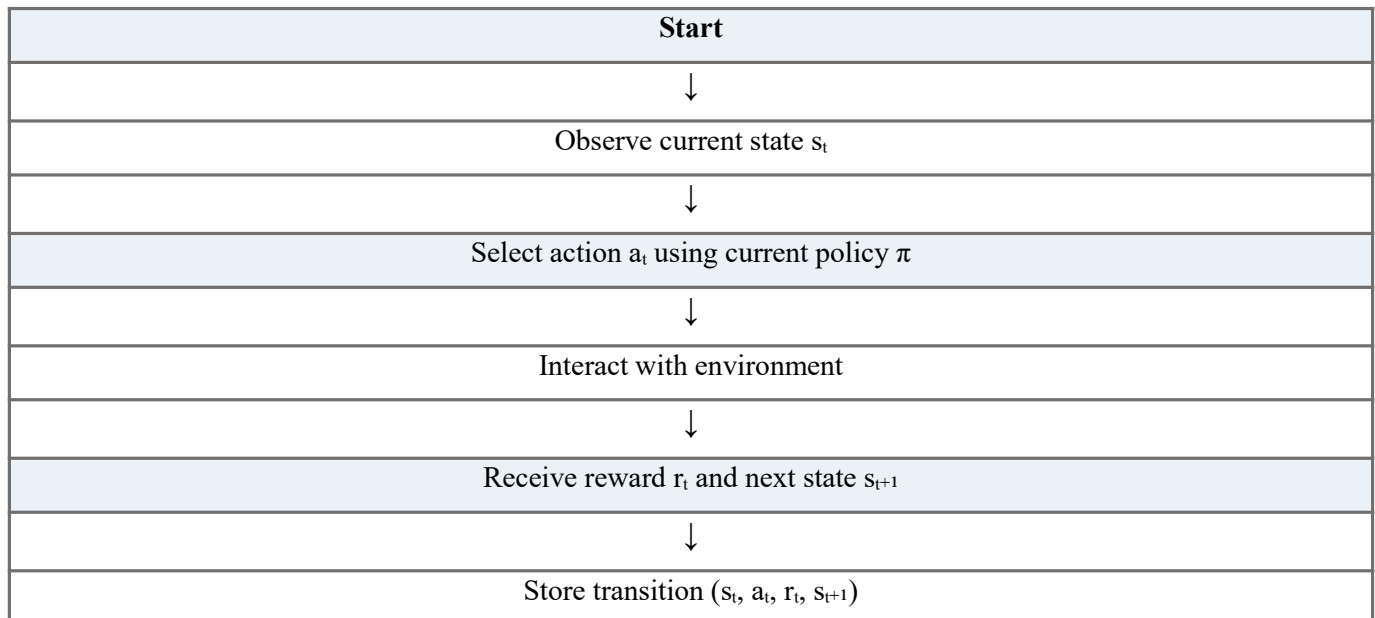
### Design and Develop Model

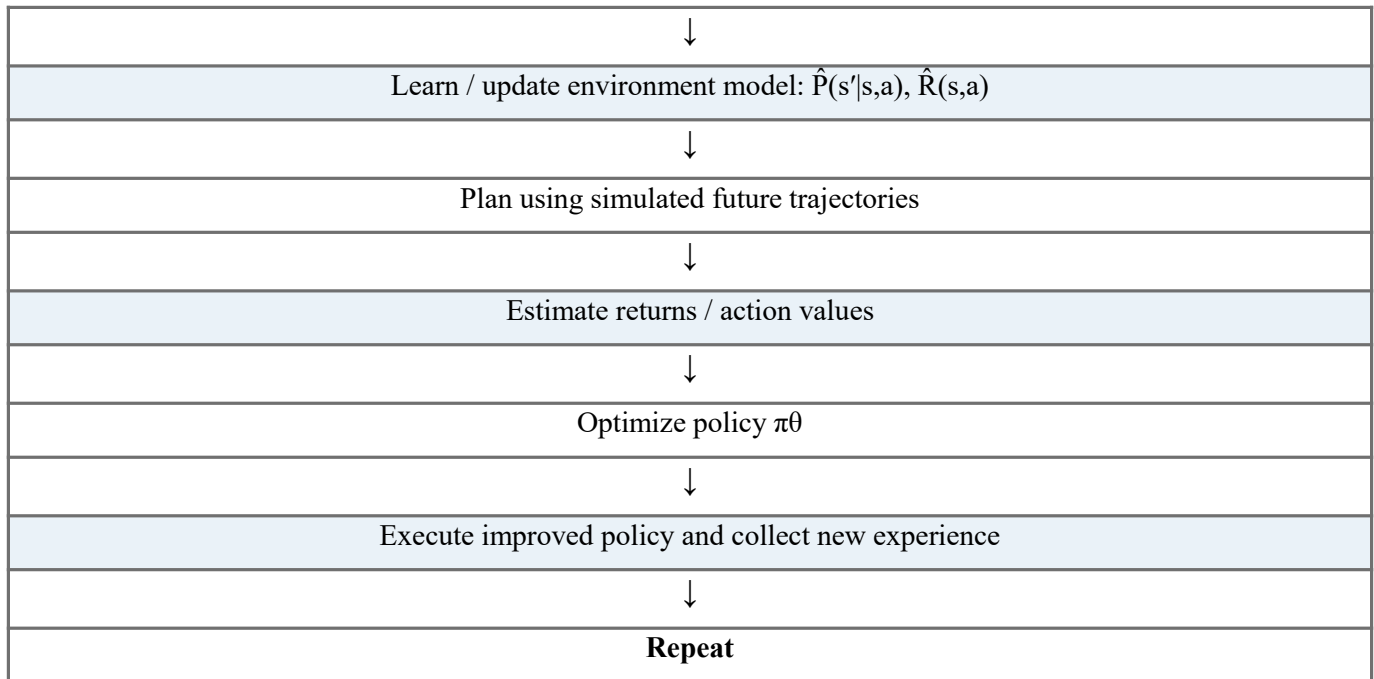
**DURATION: 4 HRS (2 Sessions)**

### Model-Based RL Approach

**1. Draw a flowchart illustrating the complete workflow of a Model-Based Reinforcement Learning agent from environment interaction to policy optimization.**

**Answer:**





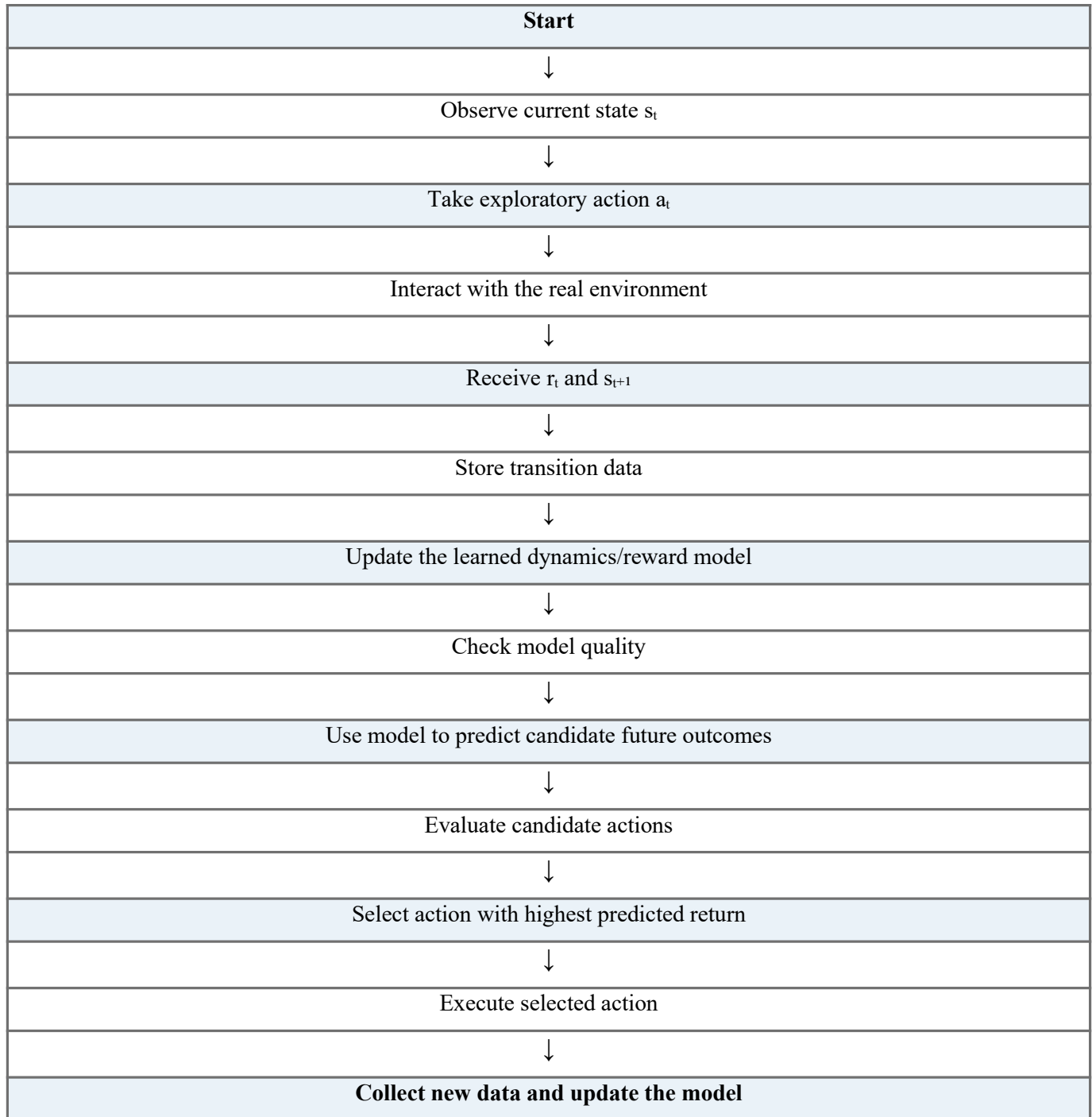
**2. Draw a concept map showing the relationship between the Environment, Model, Planning Module, Policy, and Reward in Model-Based Reinforcement Learning.**

**Answer:**

| MODEL-BASED REINFORCEMENT LEARNING |                                                                      |
|------------------------------------|----------------------------------------------------------------------|
|                                    |                                                                      |
| <b>Environment</b>                 | Provides states, transitions and rewards; receives selected actions. |
| <b>Model</b>                       | Learns/predicts transition dynamics and rewards from experience.     |
| <b>Planning Module</b>             | Uses the learned model to simulate futures and compare actions.      |
| <b>Reward</b>                      | Measures the quality of actions and provides the learning objective. |
| <b>Policy</b>                      | Selects actions and is improved using planning/value information.    |
| <b>Main loop</b>                   | Environment → experience → model → planning → policy → environment   |

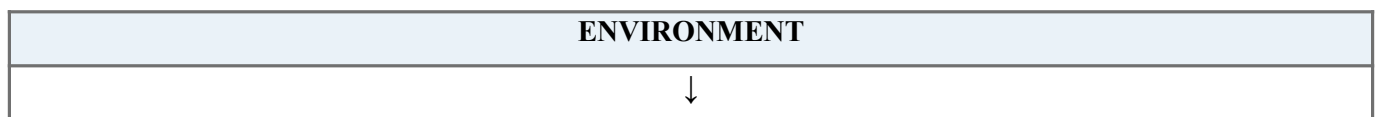
**3. Design a flowchart explaining how an RL agent learns an environment model before selecting an optimal action.**

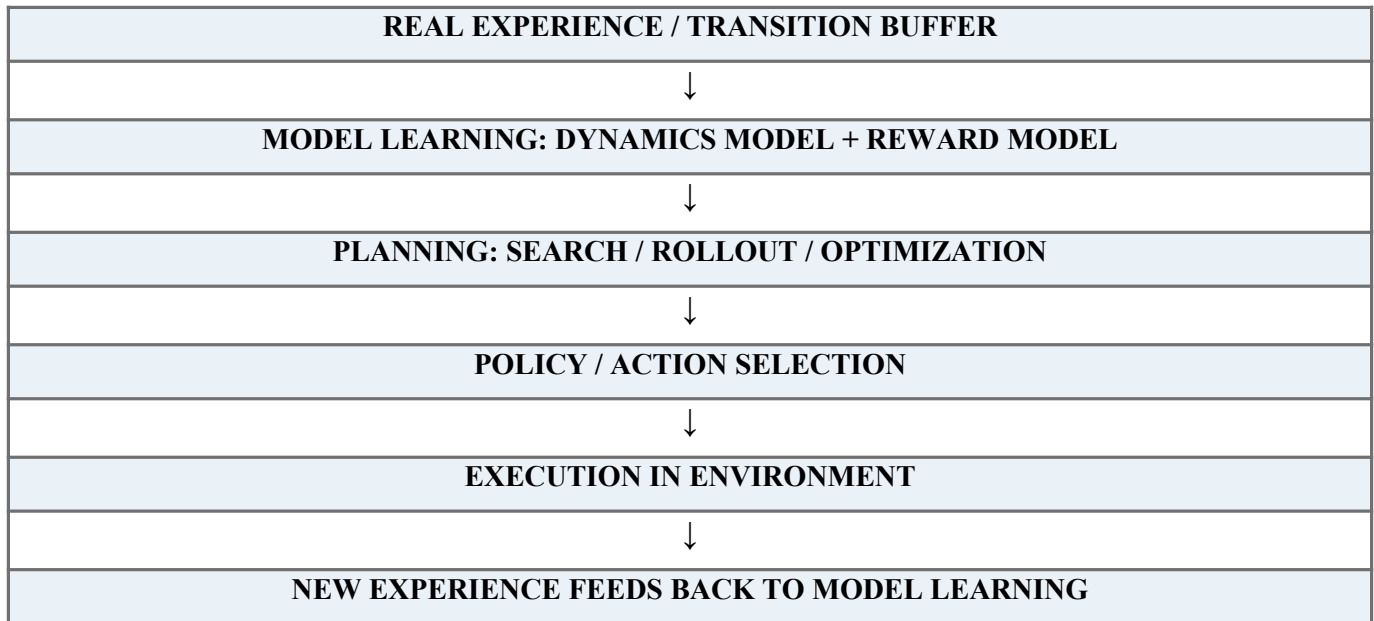
**Answer:**



**4. Draw the architecture of a Model-Based RL system showing model learning, planning, and execution components.**

**Answer:**





**5. Prepare a concept map comparing Model-Based Reinforcement Learning and Model-Free Reinforcement Learning.**

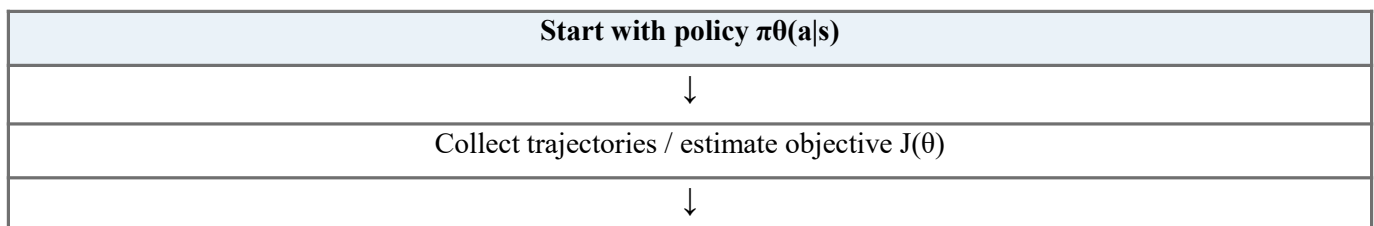
**Answer:**

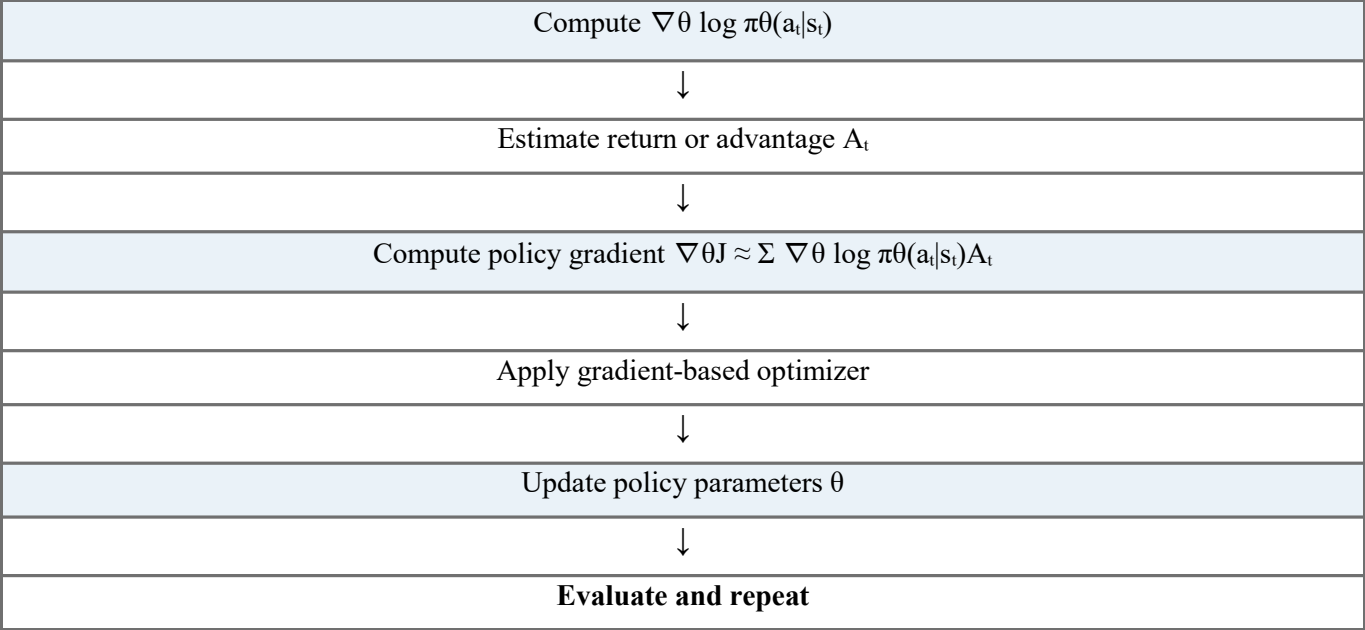
| MODEL-BASED RL                                | MODEL-FREE RL                                                |
|-----------------------------------------------|--------------------------------------------------------------|
| Learns/uses an explicit environment model.    | Learns policy/value directly from experience.                |
| Uses planning or simulated rollouts.          | Normally does not require an explicit dynamics model.        |
| Can be more sample-efficient.                 | Often needs more real interaction data.                      |
| Sensitive to model bias and prediction error. | Avoids explicit model-error problems.                        |
| More computation may be spent on planning.    | Computation is concentrated on direct policy/value learning. |

## Analytic Gradient Computation

**6. Draw a flowchart explaining the steps involved in analytic gradient computation for policy optimization.**

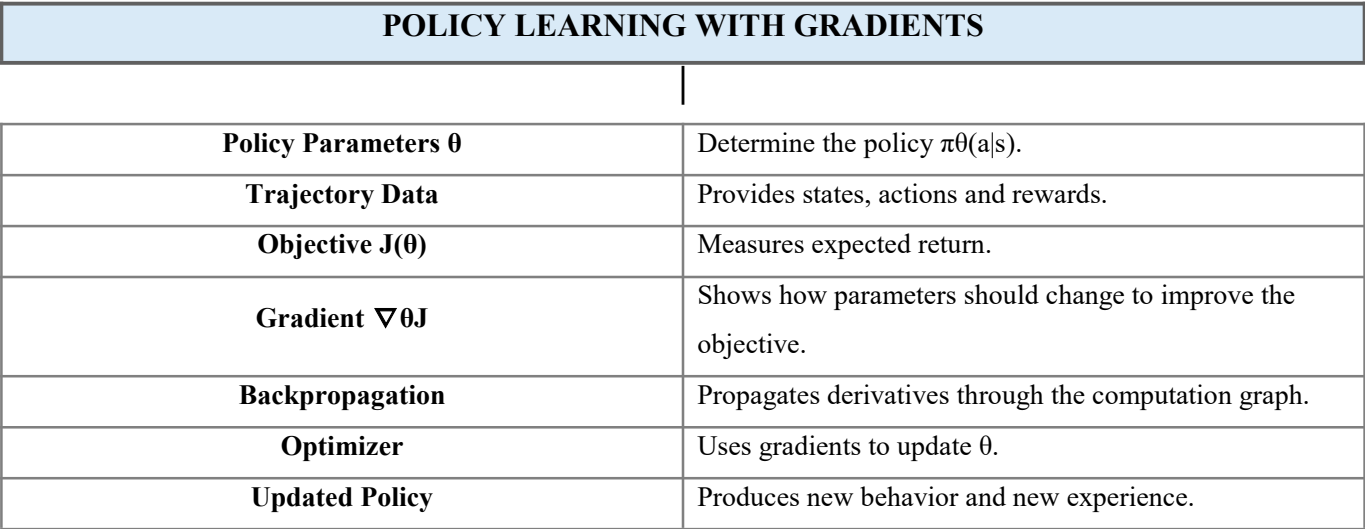
**Answer:**





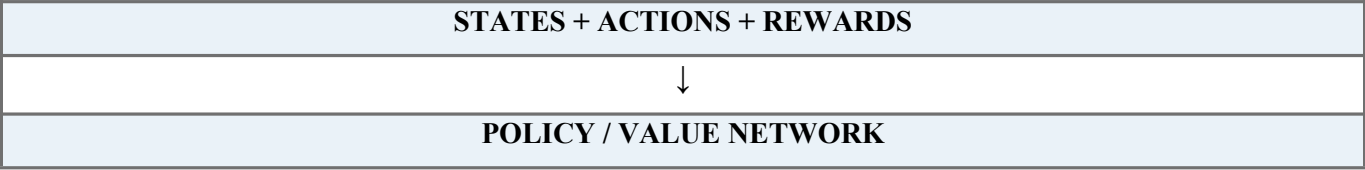
7. Design a concept map showing how gradients are computed and propagated during policy learning.

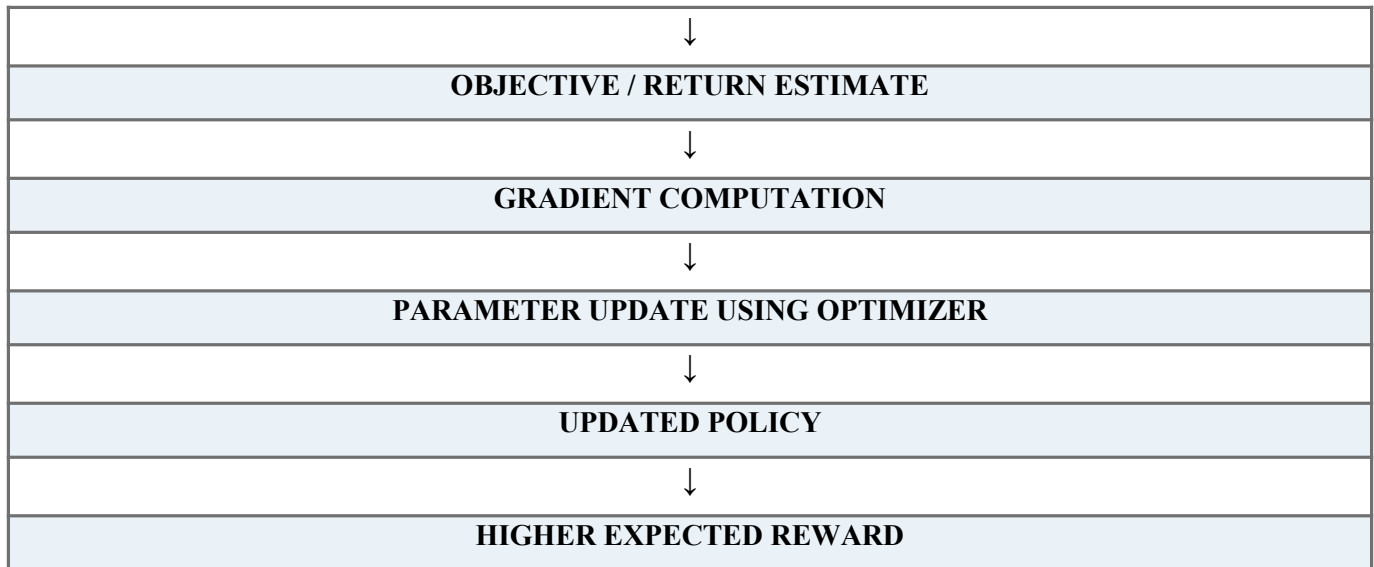
Answer:



8. Draw a block diagram illustrating gradient computation, parameter updates, and reward optimization in Reinforcement Learning.

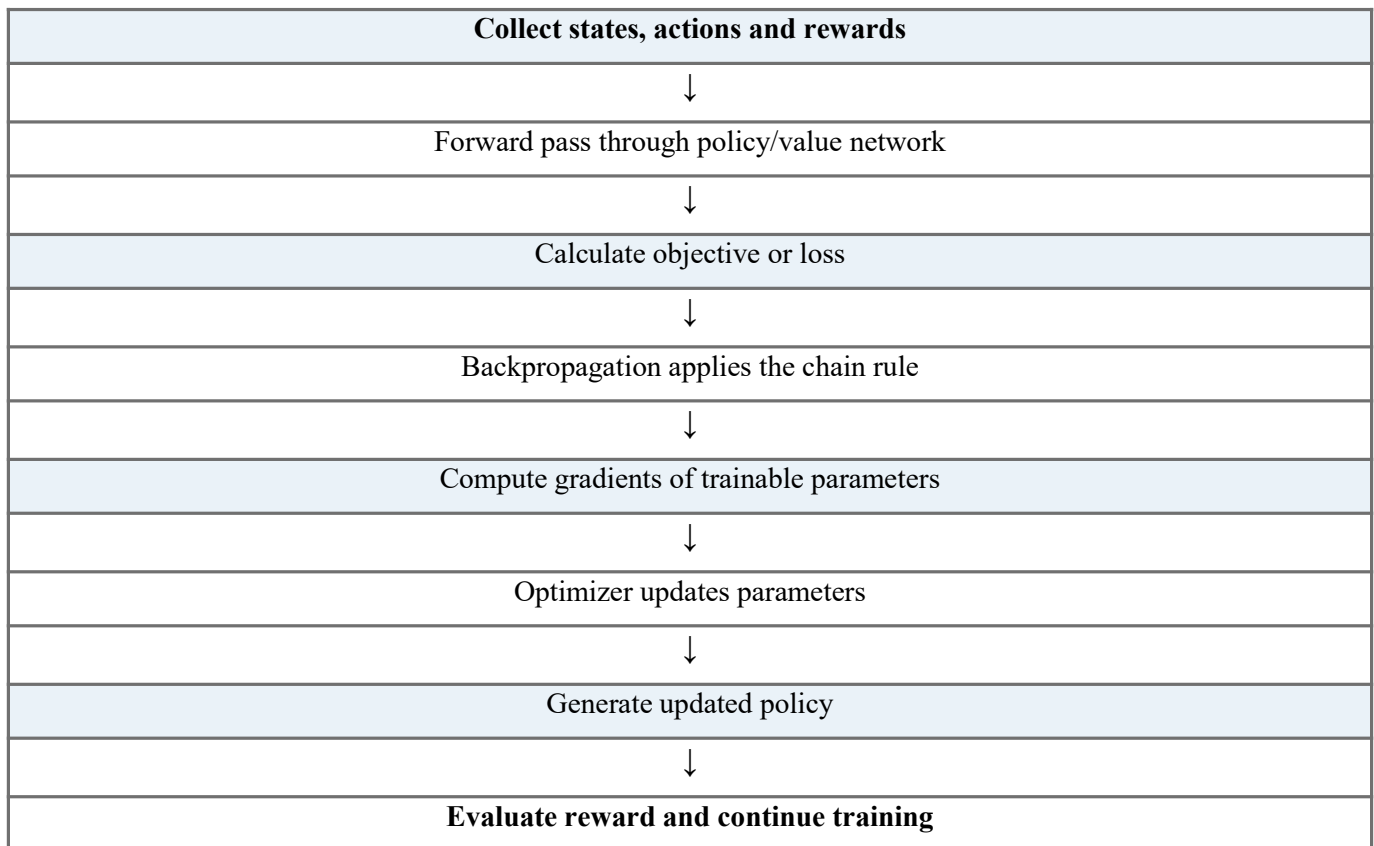
Answer:





**9. Prepare a flowchart explaining gradient-based optimization using backpropagation in Reinforcement Learning.**

**Answer:**



**10. Draw a concept map showing the relationship among policy parameters, gradients, optimization algorithms, and rewards.**

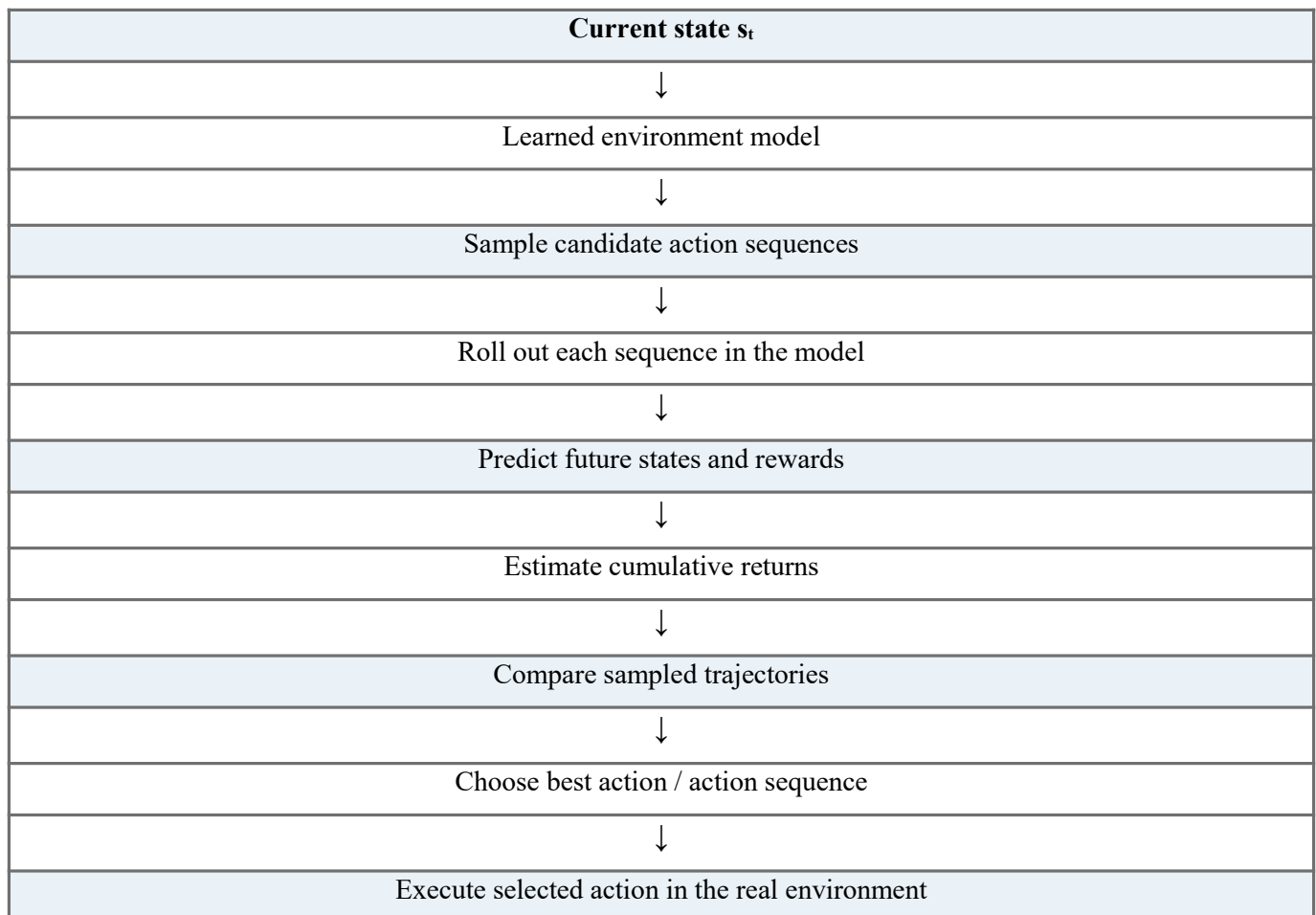
Answer:

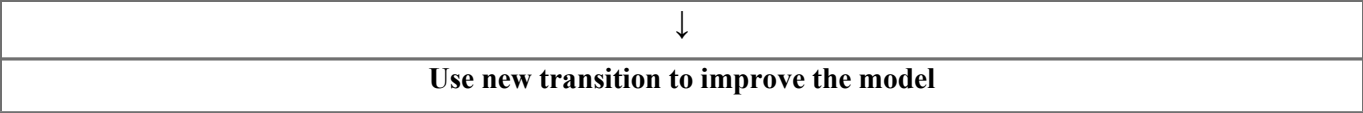
| POLICY PARAMETERS → GRADIENTS → OPTIMIZATION → REWARDS |                                                                  |
|--------------------------------------------------------|------------------------------------------------------------------|
|                                                        |                                                                  |
| Policy Parameters $\theta$                             | Control the behavior/action probabilities.                       |
| Rewards                                                | Provide feedback on action quality.                              |
| Objective $J(\theta)$                                  | Converts reward information into a learning target.              |
| Gradients $\nabla \theta J$                            | Indicate the direction and magnitude of useful parameter change. |
| Optimization Algorithm                                 | SGD/Adam applies the gradient with a learning rate.              |
| Updated $\theta$                                       | Changes the policy and influences future rewards.                |

## Sampling-Based Planning

11. Draw a flowchart illustrating the sampling-based planning process in Model-Based Reinforcement Learning.

Answer:





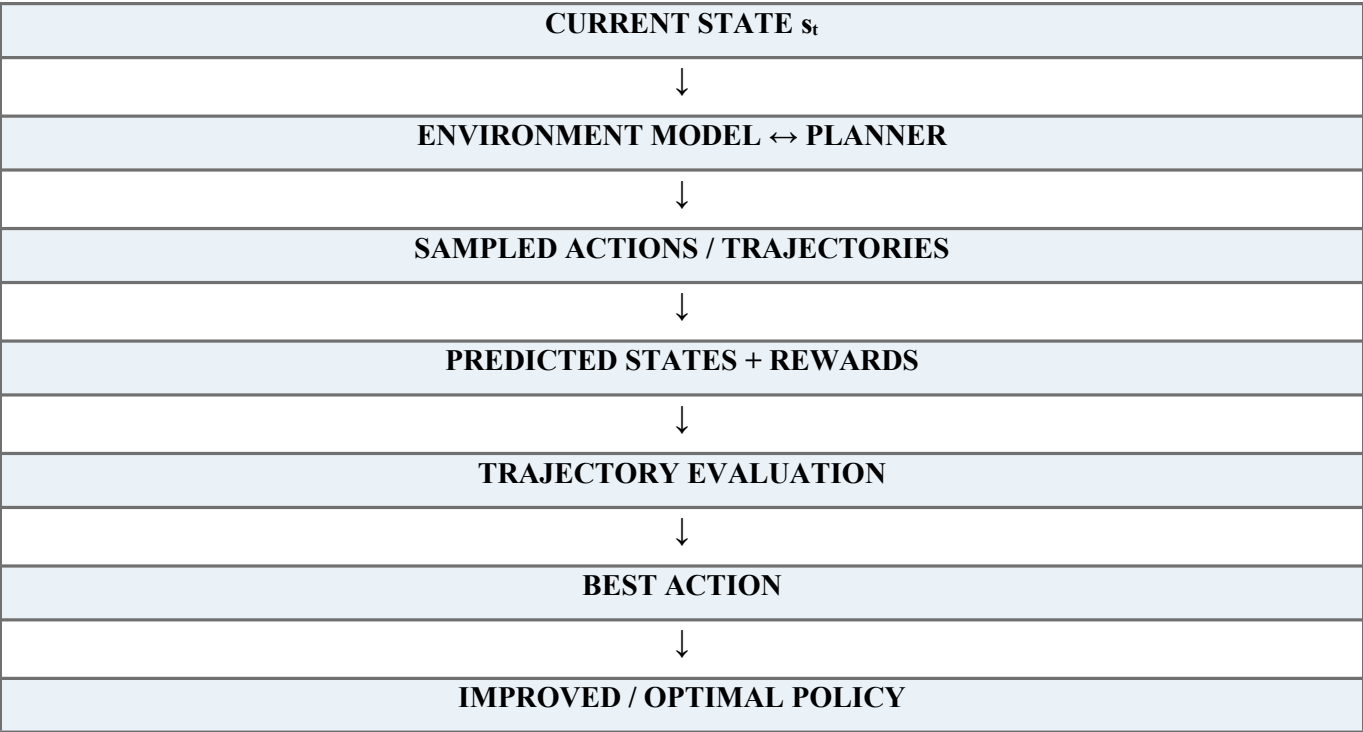
12. Prepare a concept map showing how sampled trajectories are generated and evaluated for planning.

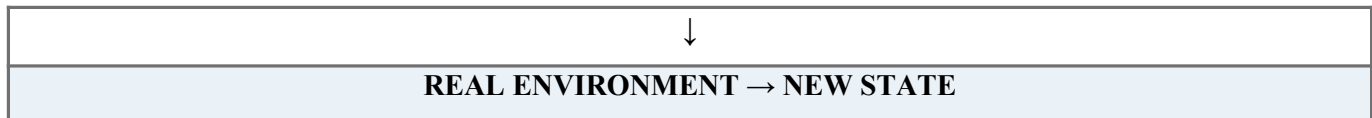
Answer:

| SAMPLED TRAJECTORY EVALUATION |                                                      |
|-------------------------------|------------------------------------------------------|
|                               |                                                      |
| Current State                 | Starting point for simulated trajectories.           |
| Sampled Actions               | Generate alternative action sequences.               |
| Environment Model             | Predicts next states and rewards.                    |
| Sampled Trajectories          | Contain predicted states, actions and rewards.       |
| Return Evaluation             | Discounted cumulative rewards score each trajectory. |
| Planner                       | Compares scores and selects the promising action.    |
| Policy                        | Uses the planning result in the real environment.    |

13. Draw a block diagram showing the interaction between the environment model, planner, sampled actions, and optimal policy.

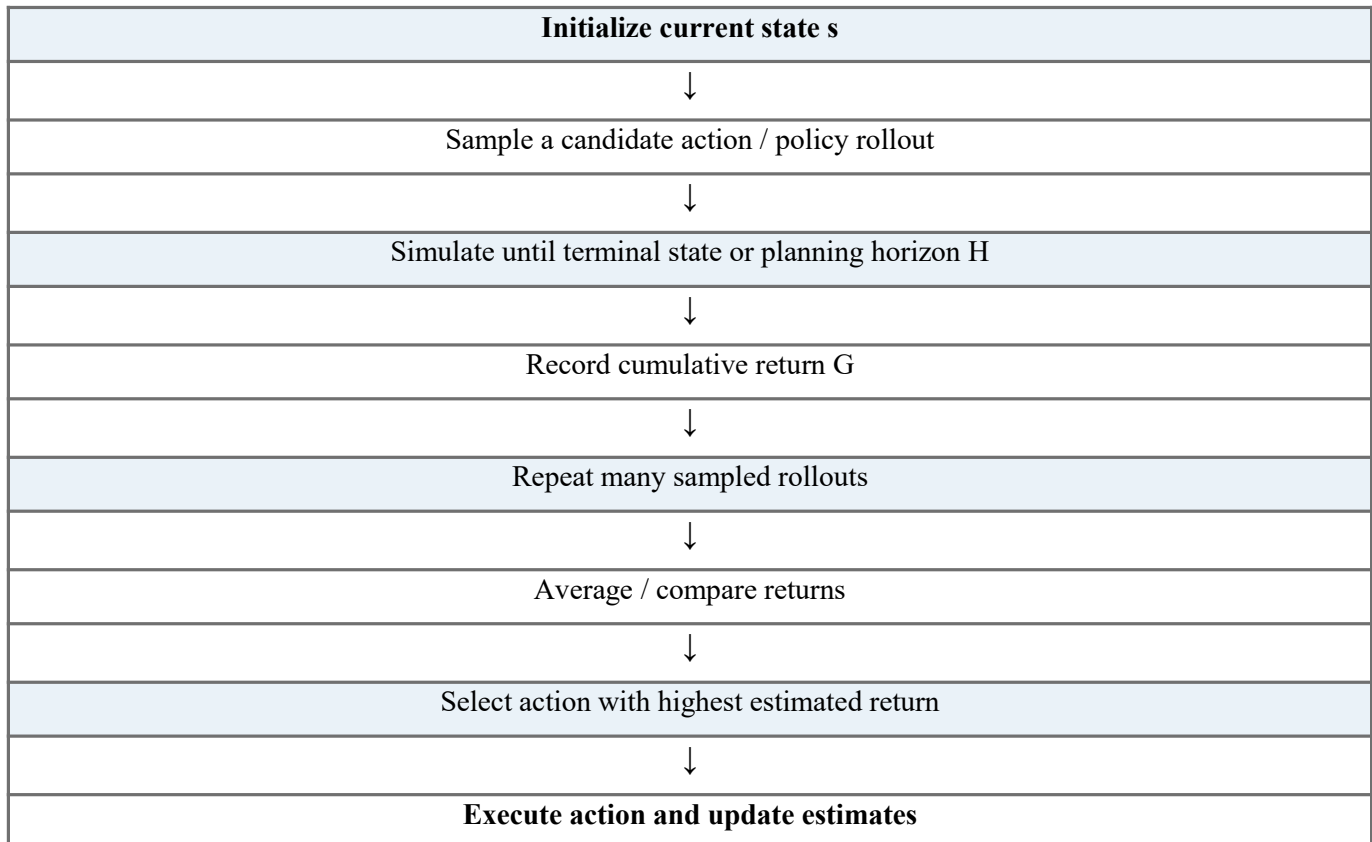
Answer:





**14. Design a flowchart explaining Monte Carlo sampling for planning in Reinforcement Learning.**

**Answer:**



**15. Draw a concept map comparing exhaustive search planning and sampling-based planning techniques.**

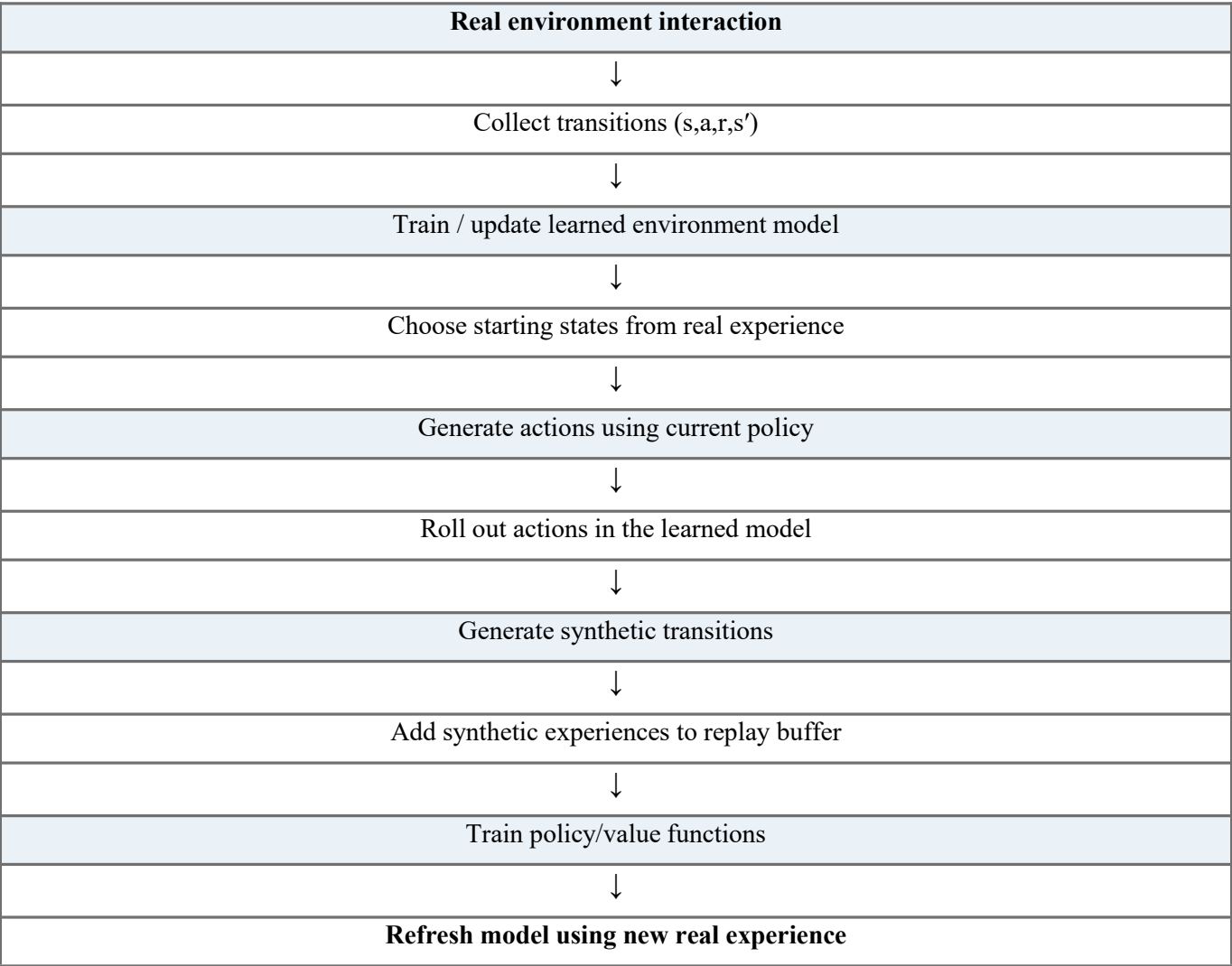
**Answer:**

| EXHAUSTIVE SEARCH                                             | SAMPLING-BASED PLANNING                           |
|---------------------------------------------------------------|---------------------------------------------------|
| Explores all feasible branches within a defined search space. | Explores a selected set of sampled trajectories.  |
| Can provide an exact best choice within the searched space.   | Provides an estimate dependent on sample quality. |
| Computational cost grows rapidly with branching factor.       | Cost can be controlled by number of samples.      |
| Poor scalability for large action spaces.                     | Generally more scalable for large spaces.         |
| High coverage but high computation.                           | Lower coverage but practical computation.         |

Model-Based Data Generation

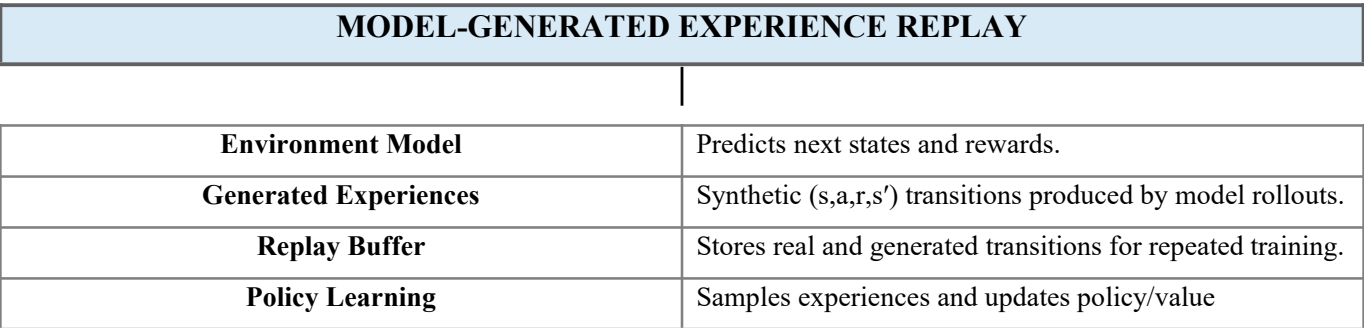
16. Draw a flowchart illustrating synthetic data generation using a learned environment model.

Answer:



17. Prepare a concept map showing the relationship among environment model, generated experiences, replay buffer, and policy learning.

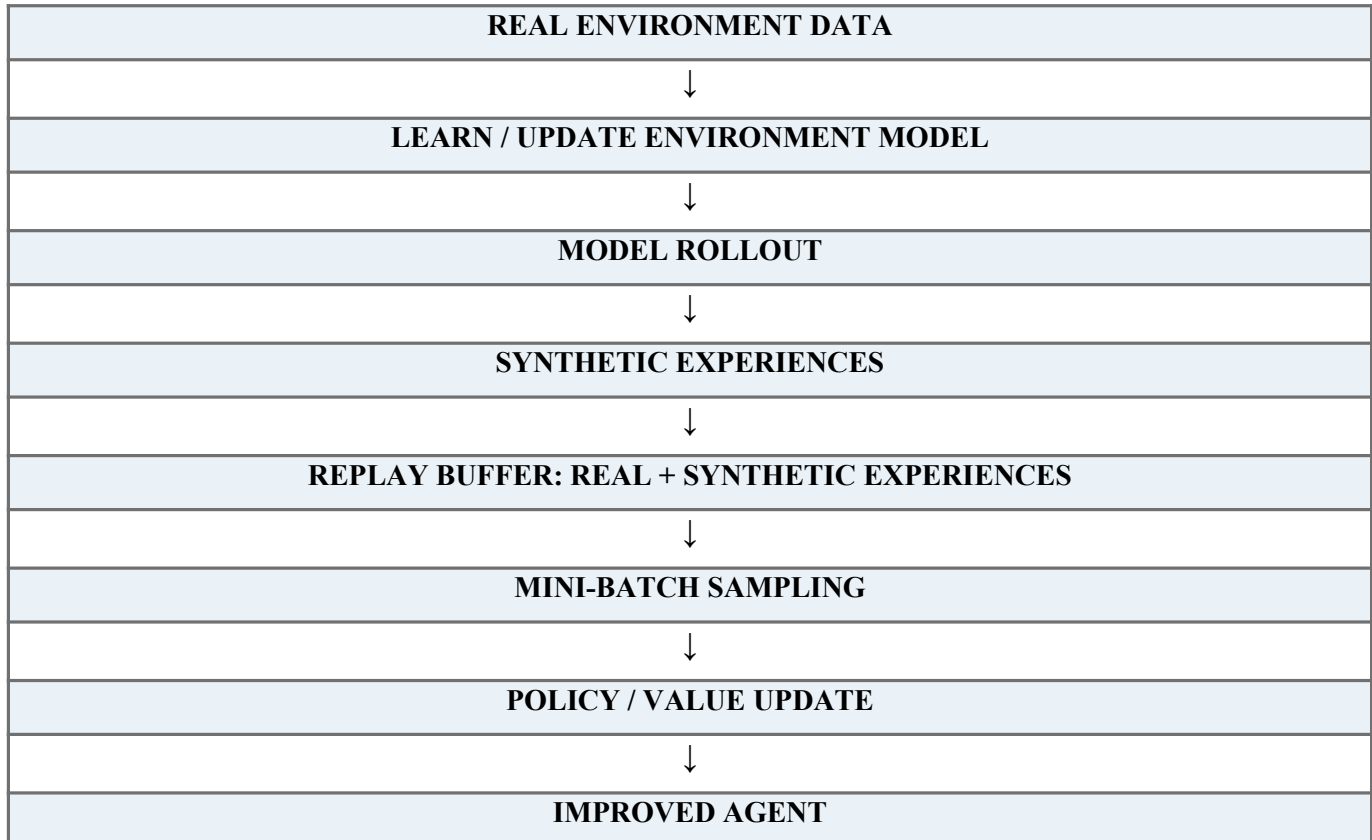
Answer:



|                      |                                                                               |
|----------------------|-------------------------------------------------------------------------------|
|                      | parameters.                                                                   |
| <b>Feedback Loop</b> | Improved policy generates new rollouts while real data corrects model errors. |

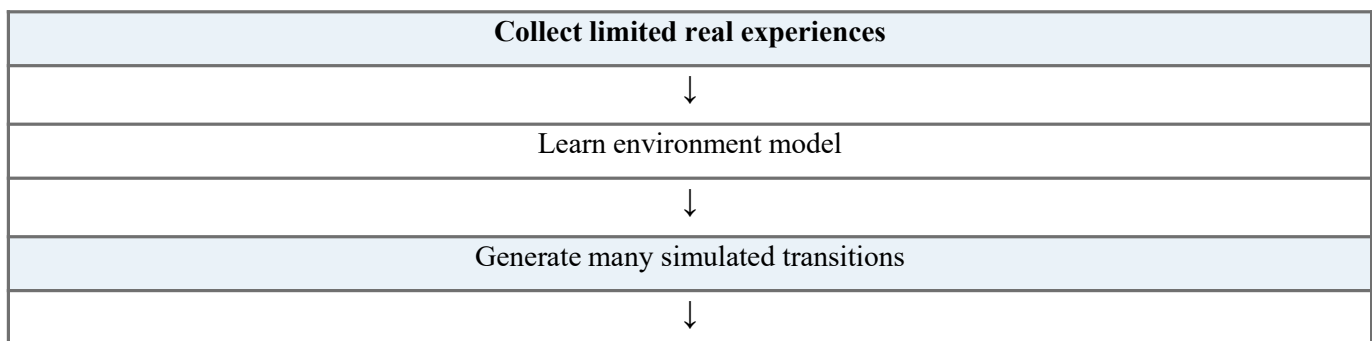
**18. Draw a block diagram explaining model-generated experience replay in Reinforcement Learning.**

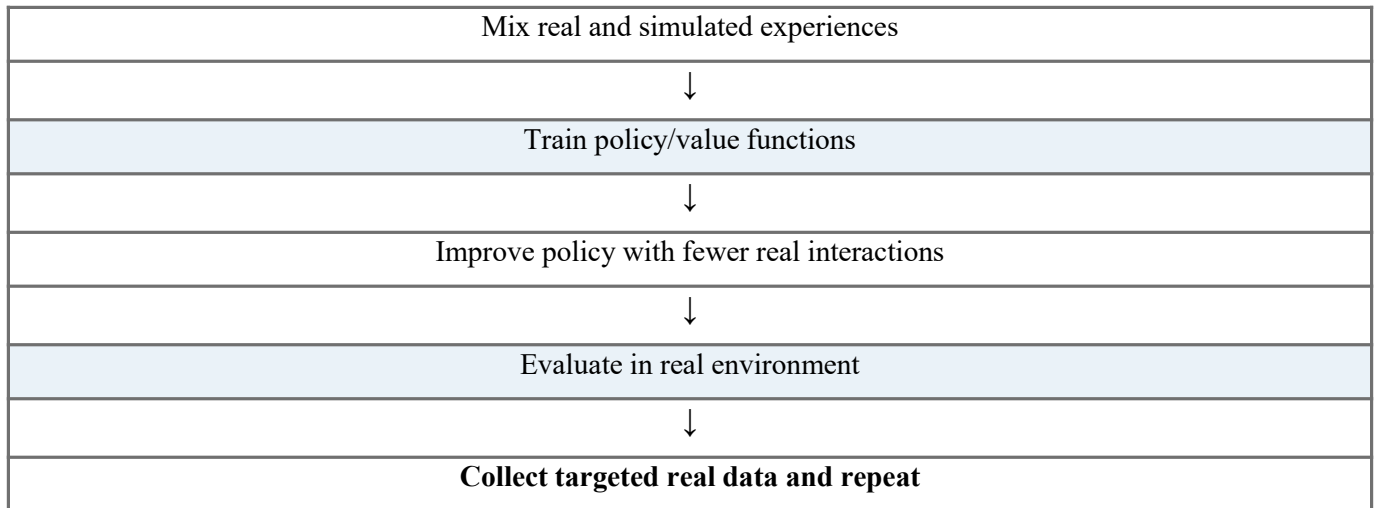
**Answer:**



**19. Design a flowchart showing how simulated experiences improve sample efficiency during training.**

**Answer:**





20. Draw a concept map illustrating the advantages and limitations of model-generated training data.

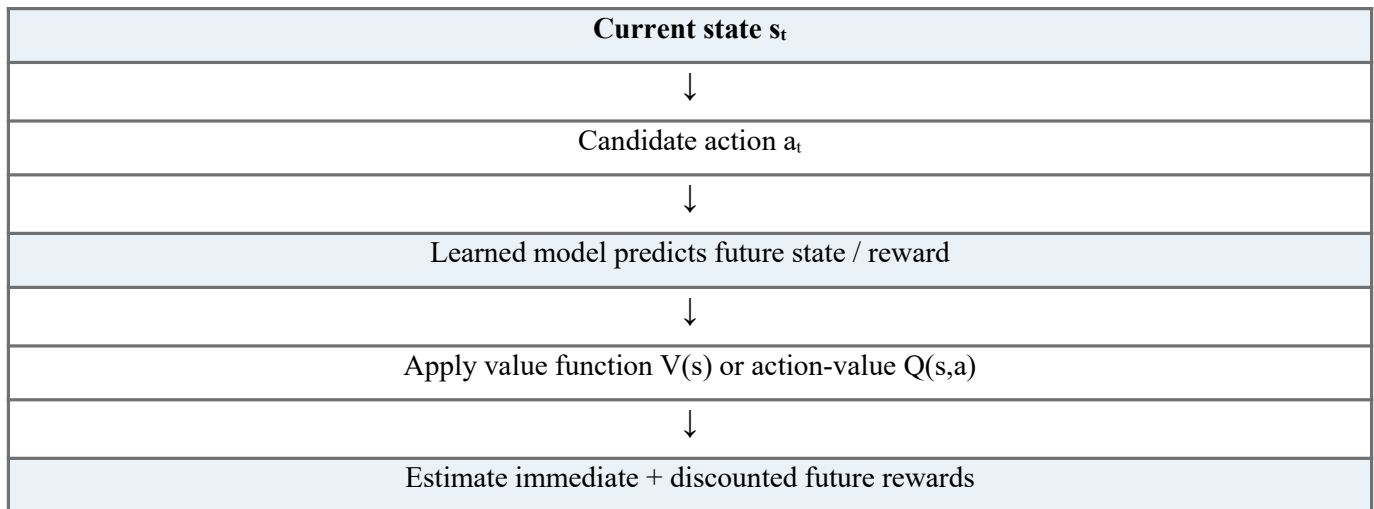
Answer:

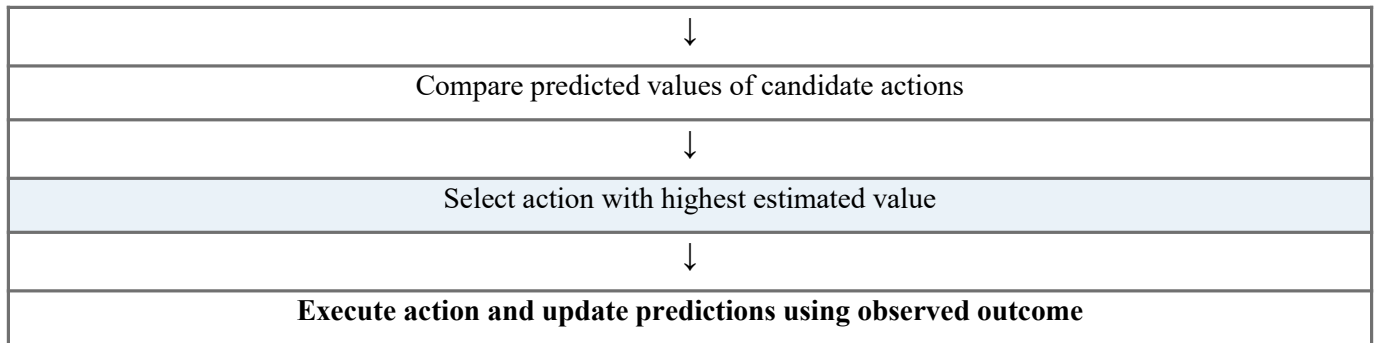
| ADVANTAGES                                           | LIMITATIONS                                            |
|------------------------------------------------------|--------------------------------------------------------|
| More training data from limited real experience.     | Model bias can create inaccurate synthetic data.       |
| Improves sample efficiency.                          | Prediction errors can compound over long rollouts.     |
| Allows safer exploration in simulation.              | Distribution shift may make synthetic data unreliable. |
| Supports planning without constant real interaction. | Model training and rollout add computational cost.     |

### Value-Equivalence Prediction and Policy Optimization

21. Draw a flowchart explaining value-equivalence prediction for estimating future rewards.

Answer:





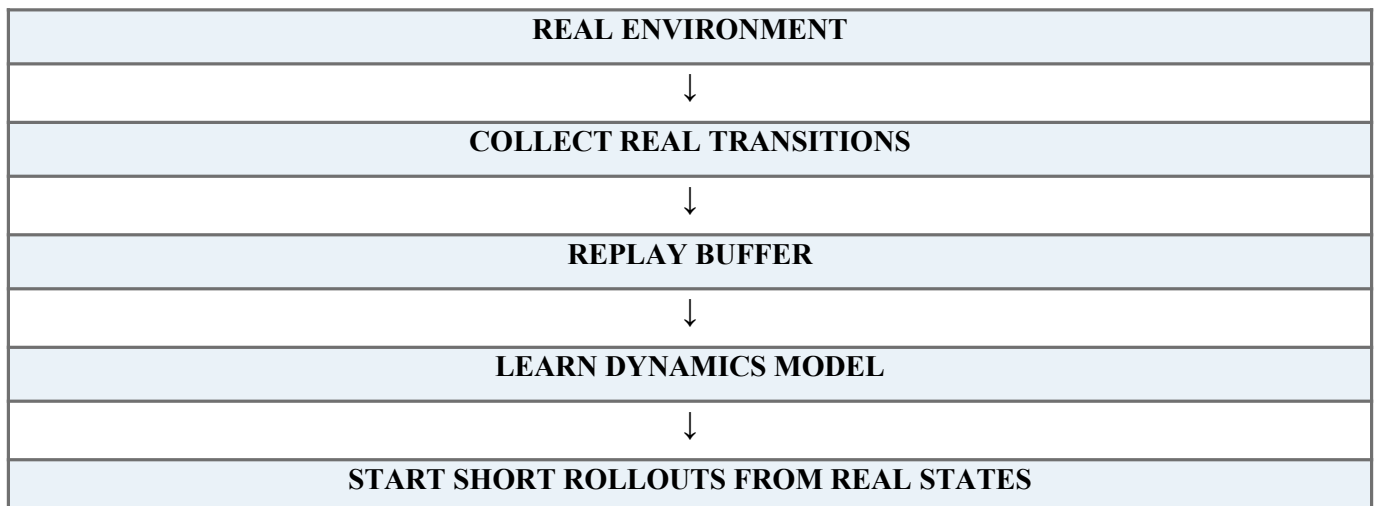
**22. Prepare a concept map showing the relationship among value functions, reward prediction, policy evaluation, and policy improvement.**

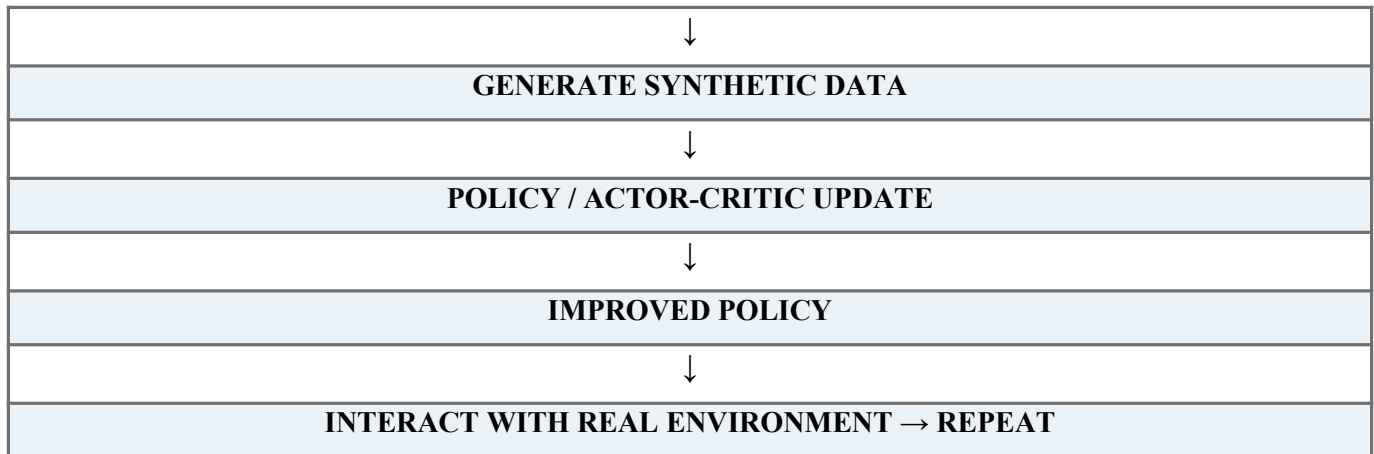
**Answer:**

| VALUE FUNCTIONS AND POLICY OPTIMIZATION |                                                              |
|-----------------------------------------|--------------------------------------------------------------|
|                                         |                                                              |
| <b>Reward Prediction</b>                | Estimates immediate reward and contributes to future return. |
| <b>Value Function <math>V(s)</math></b> | Expected return from state $s$ under a policy.               |
| <b>Action-Value <math>Q(s,a)</math></b> | Expected return for taking action $a$ in state $s$ .         |
| <b>Policy Evaluation</b>                | Uses value estimates to measure the current policy.          |
| <b>Policy Improvement</b>               | Selects actions with better estimated values.                |
| <b>Updated Policy</b>                   | Generates new experience for another evaluation cycle.       |

**23. Draw a block diagram illustrating the complete Model-Based Policy Optimization (MBPO) framework.**

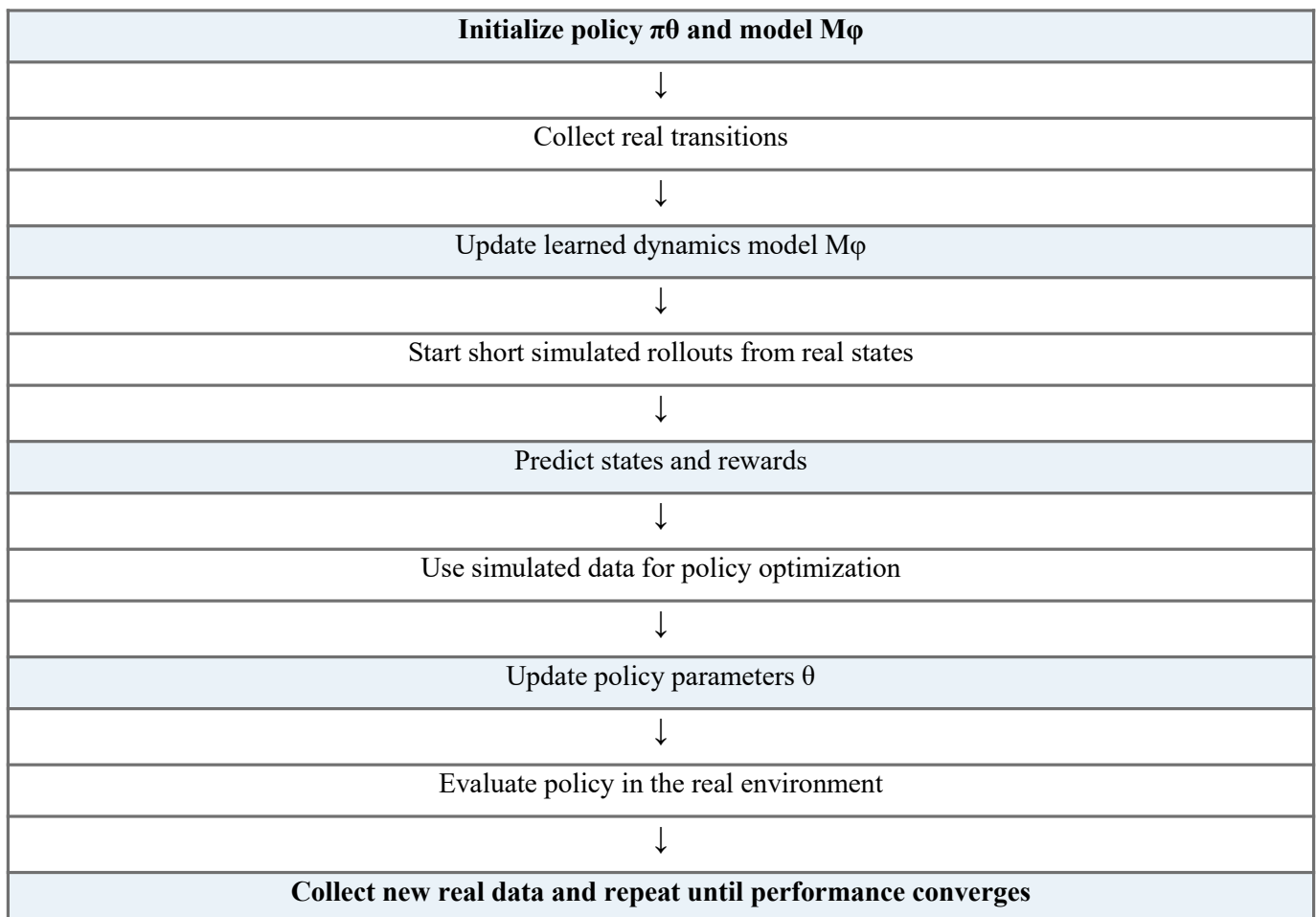
**Answer:**





**24. Design a flowchart explaining iterative policy optimization using learned environment dynamics.**

**Answer:**



**25. Draw a comprehensive concept map integrating Model-Based RL, Analytic Gradient Computation, Sampling-Based Planning, Model-Based Data Generation, Value-Equivalence**

**Prediction, and Model-Based Policy Optimization into a unified Reinforcement Learning framework.**

**Answer:**

| UNIFIED MODEL-BASED REINFORCEMENT LEARNING FRAMEWORK |                                                                                                                                 |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
|                                                      |                                                                                                                                 |
| <b>Model-Based RL</b>                                | Learns/uses environment dynamics for prediction and decision-making.                                                            |
| <b>Analytic Gradient Computation</b>                 | Computes policy gradients for parameter optimization.                                                                           |
| <b>Sampling-Based Planning</b>                       | Samples future trajectories and evaluates predicted returns.                                                                    |
| <b>Model-Based Data Generation</b>                   | Creates synthetic experiences through model rollouts.                                                                           |
| <b>Value-Equivalence Prediction</b>                  | Uses value estimates to predict future return and compare actions.                                                              |
| <b>Model-Based Policy Optimization</b>               | Combines model learning, synthetic data and policy updates.                                                                     |
| <b>Unified Loop</b>                                  | Environment → real data → learned model → planning/data/value prediction → policy optimization → improved policy → environment. |

**Code of Conduct:**

I **CHARANDEEP.N.C/192472196** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Charandeep

## RUBRICS FOR CALCULATION

| Criteria                          | Weightage | Level 4 – Exemplary                                               | Level 3 – Proficient                               | Level 2 – Developing                           | Level 1 – Beginning                               |
|-----------------------------------|-----------|-------------------------------------------------------------------|----------------------------------------------------|------------------------------------------------|---------------------------------------------------|
| Criteria                          | Weightage | Level 4 – Exemplary                                               | Level 3 – Proficient                               | Level 2 – Developing                           | Level 1 – Beginning                               |
| Understanding of Industry Problem | 20%       | Comprehensive understanding of real-world problem and constraints | Good understanding with minor gaps in requirements | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |

|                                     |     |                                                                              |                                                      |                                         |                                         |
|-------------------------------------|-----|------------------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------|-----------------------------------------|
| Application of Engineering Concepts | 25% | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts         | Incorrect or no application of concepts |
| Solution Design                     | 25% | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility | Unclear or impractical solution         |
| Use of Tools                        | 15% | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools    | Minimal or incorrect use of tools       |
| Analysis                            | 10% | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation  | Poor or no analysis                     |
| Professionalism                     | 5%  | Highly professional, well-documented, and clearly presented work             | Good documentation and presentation                  | Basic documentation with some gaps      | Poor documentation and presentation     |